

Chapter 12

HEDGING OPTIONS WITH TRANSACTION COSTS*

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Abstract

This paper studies the optimal investment problem for an investor with a HARA type utility function. We assume the investor already has an option in her portfolio, and she will setup her investment/hedging strategy using bonds and the underlying stock to maximize her utility. When there are transaction costs, the investor's optimal investment/hedging strategy can be described by three regions: the buying region, the selling region, and the no transaction region. When her portfolio falls in the buying (selling) region, she will buy (sell) enough shares of the stock to make her portfolio lie in the no transaction region (NT). When her portfolio falls in the NT region, it is optimal for the investor to make no transaction. We introduce the concept of a viscosity solution to describe the indirect utility function. A numerical scheme is proposed to compute the indirect utility function. This in turn enables the asking price for an option to be computed.

1. Introduction

In this paper we study the problem of an investor who wants to hedge her liability arising from writing an option. Hedging is central to the option pricing theory. Arbitrage pricing arguments, such as those of Black and Scholes (1973), depend on the idea that an option can be perfectly hedged using the underlying asset, so making it possible to create a port-

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folio that replicates the option payoff exactly. The standard approach dealing with option hedging is through the Black-Scholes framework, i.e., assuming in a world of frictionless markets, the option holder can hedge her position by continuously re-balancing a portfolio consisting of the underlying asset and risk-free bonds (or money market account). The number of underlying asset the investor needs to hold in order to perfectly hedge her position is computed through the Black-Scholes option pricing formula, and is often referred to as the option's *delta*. However, this hedging strategy is infeasible in the real world since transaction costs are incurred whenever the portfolio is re-balanced. Since frequent trading of the underlying stock is assumed, transaction costs will make an important impact on the hedging of options.

The problem of hedging options in the presence of proportional costs have been studied by Leland (1985), Boyle and Vorst (1991), and Toft (1996), among others. The hedging strategy considered in these studies are based on the delta hedging strategy implied from the Black-Scholes option pricing framework. They study the hedging costs when the investor tries to perfectly hedge the option under consideration. These studies ignore the rationality of the investor in the sense that the costs incurred when hedging an option in such a way may be greater than the premium she receives for the option. Moreover, the investor may not want to perfectly hedge her liability in the option position because she may prefer to take on some of the risk in exchange for a higher return for her portfolio. In other words, the hedging strategy is not based on any optimality criteria. Since no-transaction cost is one of the crucial assumptions underlying the Black-Scholes option pricing paradigm, it should be reasonable to expect that the investor would realize that the Black-Scholes option pricing formula no longer holds, and therefore the delta hedging doctrine should be abandoned.

The optimality criterion for option hedging when the market has frictions (including transaction costs) can be defined in terms of expected utility. This approach seems more appropriate than exact replication since it reflects the tradeoffs that must be made between transaction costs and risk reduction. The approach is in a paradigm similar to that of Davis and Norman (1990), and Dumas and Luciano (1991). These papers describe optimal portfolio strategies to maximize expected utility over infinite time horizon, where transaction costs are incurred whenever trades are made on the stocks. They extend earlier works by Merton (1971) and Constantinides (1986). However, while these papers are concerned with optimal portfolio strategies, they are not focused on the problem of replicating (or hedging) options by means of the underlying asset.

In this paper we study the hedging strategy for an investor who already has an option in her portfolio. The investor's optimality criterion is characterized by a HARA type utility function (the methodology can also be applied to more general cases). The approach follows the work of Davis et al (1993). However, in addition to assuming a more general utility function, our attention is focused on the hedging behavior of the investor based on her current wealth level and risk preference. Since an option's price has to be determined from an equilibrium framework when the market has frictions, we will assume that the investor already has an option in her portfolio. We will briefly discuss the minimum premium the investor should charge in order to induce her to take the liability.

The rest of this paper is organized as follows. In Section 2, we will formulate the option hedging problem, and derive the investor's wealth process. The optimal hedging strategy in absence of transaction costs is also discussed. It turns out that when the investor has a positive net wealth, she would always "over" hedge her liability in the option position in the sense that her investment in the stock is always greater than the amount of delta derived from the Black-Scholes option pricing framework, as long as the growth rate of the stock is greater than the risk-free interest rate. In Section 3, we will consider the optimal hedging/investment problem when there are transaction costs. The Hamilton-Jacobi-Bellman equation for the indirect utility function of the investment/hedging problem will be derived heuristically. In order to characterize the indirect utility function, we will introduce the concept of *viscosity solution* to the Hamilton-Jacobi-Bellman (HJB) equation, and demonstrate that the indirect utility function is indeed the unique solution in this sense. The optimal investment/hedging strategy is characterized in terms of the indirect utility function. In Section 4, we propose a numerical scheme to compute the indirect utility function based on the form of the optimal investment/hedging strategy we obtained in Section 3. Section 5 concludes.

2. An optimal hedging problem without transaction costs

We assume that there are two assets available in the market: one is a stock whose price follows a geometric Brownian motion:

$$(12.1) \quad \frac{dS}{S} = \alpha dt + \sigma dz.$$

The other asset is a risk-free bond or cash account where we assume that the interest rate is constant. In other words, one can borrow/lend at a

constant rate of $r > 0$. We write the cash account as

$$(12.2) \quad B_t = B \exp(rt),$$

where B is the initial amount in the cash account. The investor is assumed to have a utility function $u(c)$, where $u(\cdot)$ is assumed to be twice continuously differentiable and concave. We assume that the investor's objective is to maximize her expected utility from the terminal wealth at time T .

We will also assume that the investor has a short position in a European call with a strike price K and maturity time T . She balances her portfolio in a way to maximize the expected utility at time T , at which time she will liquidate her positions in the stock and fulfil her obligations in the short position of the call if it ends in the money. In other words, at time T , her final wealth is

$$(12.3) \quad W_T = B_T + y_T^* - C_T,$$

where W_T is her total wealth, B_T is the value of her investment in the risk-free account, y_t^* is the cash value of her investment in the stock, i.e., the value of her stock holding minus the transaction cost (if there is any) when she liquidates them, and C_T is the liability from the short position in the option. For an European call option, we have

$$C_T = (S_T - K)^+.$$

We derive the investor's wealth equation. Consider the following investment problem: the investor has an initial wealth of W_0 . She can invest the wealth in either the stock or the money market account. Let $h_0(t)$ and $h_1(t)$ be the amount she invests in the money market account and the number of shares she holds for the stock at time t , respectively. Her total wealth is thus

$$W(t) = h_0(t) + h_1(t)S(t).$$

Since there is no consumption before the maturity time T , we can assume that the portfolio is self-financing:

$$\begin{aligned} dW(t) &= dh_0(t) + h_1(t)dS(t) \\ &= rh_0(t)dt + h_1(t)[\alpha S(t)dt + \sigma S(t)dz_t] \\ &= [rh_0(t) + \alpha h_1(t)S(t)]dt + h_1(t)\sigma S(t)dz_t. \end{aligned}$$

Write $\pi(t)$ as the proportion of her wealth invested in the stock, i.e.,

$$\pi(t)W(t) = h_1(t)S(t),$$

then

$$(1 - \pi(t))W(t) = h_0(t),$$

and

$$\begin{aligned} dW(t) &= r(1 - \pi(t))W(t)dt + \alpha\pi(t)W(t)dt + \sigma\pi(t)W(t)dz_t \\ (12.4) \quad &= (\alpha - r)\pi W(t)dt + rW(t)dt + \sigma\pi W(t)dz_t. \end{aligned}$$

The investor's problem is to maximize her expected utility at the maturity time:

$$\max_{\pi(\cdot)} Eu(W_T),$$

and the investor's indirect utility function is defined as follows: for an initial wealth W at time t ,

$$U(t, W) = \max_{\pi(\cdot)} E_t u(W_T).$$

Let us introduce the following differential operator:

$$\mathcal{L}^\pi \equiv \frac{1}{2}\pi^2 W^2 \sigma^2 \frac{\partial^2}{\partial W^2} + [(\alpha - r)\pi W + rW] \frac{\partial}{\partial W},$$

then the Hamilton-Jacobi-Bellman (HJB hereafter) equation for the investment problem is

$$(12.5) \quad \frac{\partial U}{\partial t} + \max_{\pi} \mathcal{L}^\pi U = 0$$

with the boundary condition

$$U(T, W) = u(W), \quad \forall W \geq 0.$$

Once the indirect utility function U is found, the optimal strategy π^* is given by

$$\pi W^2 \sigma^2 \frac{\partial^2 U}{\partial W^2} + (\alpha - r)W \frac{\partial U}{\partial W} = 0,$$

or

$$(12.6) \quad \pi^* = -\frac{(\alpha - r)U_W}{W\sigma^2 U_{WW}}.$$

In order to solve the HJB equation, let us assume that the utility function is in the HARA class (hyperbolic absolute risk aversion):

$$(12.7) \quad u(W) = \frac{W^\gamma}{\gamma}, \quad 0 < \gamma < 1.$$

This type of utility function is widely used in the optimal investment/consumption problems. Note that the relative risk aversion is given by

$$-W \frac{u''(W)}{u'(W)} = 1 - \gamma.$$

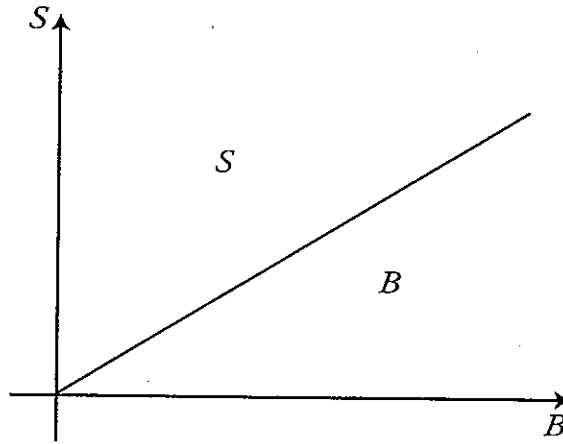


Figure 12.1. The case with no transaction costs

2.1 Optimal investment problem without options

In this case, the boundary condition of the HJB equation is thus

$$U(T, W) = \frac{W^\gamma}{\gamma}, \quad \forall W \geq 0.$$

Under these assumptions, the indirect utility function can be found as (see Appendix):

$$U(t, W) = g(t) \frac{W^\gamma}{\gamma},$$

where

$$(12.8) \quad g(t) = e^{\nu r(T-t)}$$

and

$$\nu = \frac{1}{2} \left(\frac{\alpha - r}{\sigma} \right)^2 \frac{1}{1 - \gamma} + r.$$

From (12.6), the optimal investment policy is given by

$$\pi^* = \frac{\alpha - r}{\sigma^2(1 - \gamma)}.$$

In this case, the investor's optimal investment strategy is to keep a fixed proportion π^* of her wealth in the stock. This means that,

optimally, the investor acts in such a way that her portfolio holding is always on the line

$$S = \frac{\pi^*}{1 - \pi^*} B.$$

which is sometimes referred to as the *Merton line*.¹ It is worth mentioning that when $\alpha > r$, the investor always invests some portion of her wealth in the stock. Moreover, for this choice of the utility function, the proportion of the investor's wealth invested in the stock is independent of her wealth level.

This result is shown in Figure 12.1. When the portfolio falls in the region $\mathcal{B}$, the investor shall purchase enough shares of stock to keep her portfolio on the Merton line. Similarly, when her portfolio is in the region $\mathcal{S}$, she should sell some shares of the stock she is holding to move her portfolio on the Merton line. We will refer $\mathcal{B}$ and $\mathcal{S}$ as the *buying region* and *selling region* respectively.

2.2 When there is a short position in option

We now consider the case when there is a short position of European call already in her portfolio. We want to answer the following question: based on her wealth level and preference, what is her best investment strategy? Notice that the investor might not want to perfectly hedge her position in the option since she might prefer to have some risk in her portfolio in order to achieve a higher return. This is illustrated by the previous case when there is no option in her portfolio.

The terminal wealth in this case is

$$(12.9) \quad \bar{W}_T = W_T - (S_T - K)^+.$$

The optimal investment/hedging strategy can be constructed in the following way. It is easy to show that for any time $t \leq T$ and wealth level $W \geq 0$, the indirect utility function is given by

$$(12.10) \quad \begin{aligned} \bar{U}(t, W) &= U(t, W - f(t, S)) \\ &= g(t) \frac{[W - f(t, S)]^\gamma}{\gamma}, \end{aligned}$$

where $g(t)$ is given by (12.8) and $f(t, S)$ is the price of the call option under the Black-Scholes framework, i.e.,

$$f(t, S) = SN(d_1) - Ke^{-r(T-t)}N(d_2),$$

with

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = d_1 - \sigma\sqrt{T - t}.$$

For a proof of this conclusion, see the Appendix.

Equation (12.10) suggests that the optimal strategy for the investor is to delta-hedge her short position in the European call, which needs an amount $f(0, S)$ initially, and then manage her remaining wealth optimally as before. While this is true, it is important to look at what her overall investment/hedging strategy is based on her wealth and risk preference.

In order to find the optimal investment/hedging strategy, recall that the stock price and the wealth processes are given by (12.1) and (12.4), respectively. The indirect utility is defined as

$$U(t, S, W) \equiv \max_{\pi} Eu(\bar{W}_T),$$

where $\bar{W}_T$ is given by (12.9). Unlike the previous case, we now have to include the current stock price as one of the variables for the indirect utility function. Similarly, let

$$\begin{aligned} \bar{\mathcal{L}}^{\pi} \equiv & \frac{1}{2} \pi^2 W^2 \sigma^2 \frac{\partial^2}{\partial W^2} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2}{\partial S^2} + \pi W S \sigma^2 \frac{\partial^2}{\partial S \partial W} \\ & + [(\alpha - r) \pi W + r W] \frac{\partial}{\partial W} + \alpha S \frac{\partial}{\partial S}, \end{aligned}$$

then the HJB equation for this problem becomes

$$U_t + \max_{\pi} \bar{\mathcal{L}}^{\pi} U = 0,$$

with the boundary condition given by

$$U(T, S, W) = \frac{(W - (S - K)^+)^{\gamma}}{\gamma}.$$

The optimal strategy is given by

$$\pi W^2 \sigma^2 U_{WW} + W S \sigma^2 U_{SW} + (\alpha - r) W U_W = 0,$$

or

$$(12.11) \quad \pi^* = -\frac{\alpha - r}{\sigma^2 W} \cdot \frac{U_W}{U_{WW}} - \frac{S}{W} \cdot \frac{U_{SW}}{U_{WW}}.$$

Given the solution in (12.10), we have

$$\begin{aligned} U_W &= g(t) \gamma [W - f(t, S)]^{\gamma-1}, \\ U_{WW} &= g(t) \gamma (\gamma - 1) [W - f]^{\gamma-2}, \\ U_S &= -g(t) \gamma [W - f]^{\gamma-1} \cdot f_S, \\ U_{SW} &= -\gamma (\gamma - 1) g(t) [W - f]^{\gamma-2} \cdot f_S. \end{aligned}$$

The optimal investment/hedging strategy can be written as

$$(12.12) \quad \pi^* = \frac{\alpha - r}{\sigma^2(1 - \gamma)} \cdot \frac{W - f}{W} + \frac{S}{W} \cdot f_S.$$

It is interesting to notice that in this case, the proportion of the investor's wealth invested in the stock depends on her wealth level.

The wealth that is invested in the stock is given by

$$\pi^* W = \frac{\alpha - r}{\sigma^2(1 - \gamma)} \cdot (W - f) + S \cdot f_S,$$

which implies that she will perfectly hedge her liability in the option only if her initial wealth equals the premium she receives from the short position in the option. Assuming $\alpha > r$ and that the investor has more wealth than just the option premium, the investor will always invest more than the amount that is necessary for the delta hedging strategy for the option derived from the Black-Scholes model.

3. The case when there are transaction costs

Now we consider the case when transaction costs are incurred when one buys/sells stocks. We still assume that there are two assets available: one is the risky asset, or a stock, whose price is described by a geometric Brownian motion

$$\frac{dS}{S} = \alpha dt + \sigma dz,$$

and the other asset is a money market account (or risk-free bond) with a risk-free interest of r .

As before, the investor's wealth is distributed among investments in both the stock and the money market account. Moreover, we assume that a proportional cost is incurred each time a transaction of buying/selling of the stock is made. More specifically, in order to purchase (sell) one share of the stock, he will pay (receive)

$$(1 + \lambda)S \quad ((1 - \mu)S).$$

Her wealth can be described by

$$(12.13) \quad dW_t = rW_t dt - (1 + \lambda)S_t dL_t + (1 - \mu)S_t dM_t,$$

where L_t and M_t are the cumulative amount of shares of the stock the investor has bought and sold up to time t , respectively. For technical reasons, we assume

- L and M are adapted to the information generated by the price process S ;

- $L(\cdot)$ and $M(\cdot)$ are right continuous and have left limits;
- $L(0-) = M(0-) = 0$.

The total number of shares the investor holds at time t is given by

$$y(t) = L(t) - M(t).$$

An investment/hedging strategy is described by the pair (L, M) .

3.1 The optimal investment problem without options

As before, we assume that the investor has a utility function given by (12.7) and she choose the investment/hedging strategy (L, M) to maximize her expected utility at time T :²

$$\max_{(L, M)} Eu(W_T).$$

In order to derive the optimal investment strategy, we define the indirect utility function as

$$U(t, S, y, W) \equiv \max_{(L, M)} E_t(u(W(T))),$$

where

- t is the current time;
- S is the spot price of the stock at time t ;
- y is the net position of the investor's position in the stock;
- W is the investor's net wealth (or cash value) at time t .

When the utility function is assumed to be in the form (12.7), it is easy to show the following: for the special choice of the utility function u , for any $\rho > 0$, we have

$$U(t, \rho S, y, \rho W) = \rho^\gamma U(t, S, y, W).$$

The HJB equation

For any positive constants C_1 and C_2 , consider the subset $\mathcal{S}_{C_1, C_2}$ of trading strategies in the following form

$$L(t) = \int_0^t l(s)ds, \quad M(t) = \int_0^t m(s)ds$$

such that

$$0 \leq l(s) \leq C_1, \quad 0 \leq m(s) \leq C_2.$$

The wealth process can be written as

$$\begin{aligned} dW &= [rW - (1 + \lambda)l(t)S + (1 - \mu)m(t)S] dt, \\ dy &= [l(t) - m(t)] dt, \\ \frac{dS}{S} &= \alpha dt + \sigma dz. \end{aligned}$$

Define

$$U^{C_1, C_2}(t, S, y, W) \equiv \max_{(L, M) \in \mathcal{S}_{C_1, C_2}} E_t(u(W(T))),$$

then the investment problem reduces to a case similar to the one considered in Section 2. As a result, the HJB equation for $U^{C_1, C_2}(t, S, y, W)$ take the following form:

$$(12.14) \quad \frac{\partial U}{\partial t} + \max_{0 \leq l \leq C_1, 0 \leq m \leq C_2} \mathcal{L}^{l, m} U = 0,$$

where

$$\begin{aligned} \mathcal{L}^{l, m} &= \frac{1}{2} \sigma^2 S^2 \frac{\partial^2}{\partial S^2} + \alpha S \frac{\partial}{\partial S} \\ &\quad + [rW - (1 + \lambda)lS + (1 - \mu)mS] \frac{\partial}{\partial W} + (l - m) \frac{\partial}{\partial y} \\ &= \frac{1}{2} \sigma^2 S^2 \frac{\partial^2}{\partial S^2} + \alpha S \frac{\partial}{\partial S} + rW \frac{\partial}{\partial W} \\ &\quad + \left[\frac{\partial}{\partial y} - (1 + \lambda)S \frac{\partial}{\partial W} \right] l + \left[-\frac{\partial}{\partial y} + (1 - \mu)S \frac{\partial}{\partial W} \right] m. \end{aligned}$$

From the HJB equation (12.14), we can see that the optimal investment strategy is given by

$$\begin{aligned} l^* &= \begin{cases} C_1 & \text{if } \frac{\partial U^{C_1, C_2}}{\partial y} - (1 + \lambda)S \frac{\partial U^{C_1, C_2}}{\partial W} > 0, \\ 0 & \text{otherwise;} \end{cases} \\ m^* &= \begin{cases} C_2 & \text{if } -\frac{\partial U^{C_1, C_2}}{\partial y} + (1 - \mu)S \frac{\partial U^{C_1, C_2}}{\partial W} > 0, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Letting $C_1, C_2 \rightarrow \infty$, we can expect that

$$U^{C_1, C_2} \rightarrow U,$$

$$\begin{aligned}
\frac{\partial U^{C_1, C_2}}{\partial W} &\rightarrow \frac{\partial U}{\partial W}, \\
\frac{\partial U^{C_1, C_2}}{\partial y} &\rightarrow \frac{\partial U}{\partial y}, \\
\frac{\partial U^{C_1, C_2}}{\partial S} &\rightarrow \frac{\partial U}{\partial S}, \\
\frac{\partial^2 U^{C_1, C_2}}{\partial S^2} &\rightarrow \frac{\partial^2 U}{\partial S^2}.
\end{aligned}$$

As a result, the HJB takes the form of the so called *variational inequality*:

$$(12.15) \quad \max \left\{ \frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W}, -\frac{\partial U}{\partial y} + (1 - \mu)S \frac{\partial U}{\partial W}, \right. \\
\left. \frac{\partial U}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + \alpha S \frac{\partial U}{\partial S} + rW \frac{\partial U}{\partial W} \right\} = 0.$$

From the derivation of this equation, we can see that the (S, y, W) space can be divided into three regions:

NT: This region is defined by

$$\begin{aligned}
\frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W} &\leq 0, \\
-\frac{\partial U}{\partial y} + (1 - \mu)S \frac{\partial U}{\partial W} &\leq 0, \\
\frac{\partial U}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + \alpha S \frac{\partial U}{\partial S} + rW \frac{\partial U}{\partial W} &= 0.
\end{aligned}$$

S: This region is defined by

$$\begin{aligned}
\frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W} &= 0, \\
-\frac{\partial U}{\partial y} + (1 - \mu)S \frac{\partial U}{\partial W} &< 0, \\
\frac{\partial U}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + \alpha S \frac{\partial U}{\partial S} + rW \frac{\partial U}{\partial W} &\leq 0.
\end{aligned}$$

B: This region is defined by

$$\begin{aligned}
\frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W} &< 0, \\
-\frac{\partial U}{\partial y} + (1 - \mu)S \frac{\partial U}{\partial W} &= 0, \\
\frac{\partial U}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 U}{\partial S^2} + \alpha S \frac{\partial U}{\partial S} + rW \frac{\partial U}{\partial W} &\leq 0.
\end{aligned}$$

The region $\mathcal{S}$ is called the *selling region*. In order to see this, define a function in the following form: for $h > 0$,

$$g(h) = U(t, S, y + h, W - (1 + \lambda)Sh),$$

which represents the utility change when h number of shares of the stock are purchased at time t . If $(S, y, W) \in \mathcal{S}$, then

$$g'(h) = \frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W} = 0,$$

in other words, the investor will immediately sell enough shares of the stock so that her portfolio will lie on $\partial\mathcal{S}$.

Similarly, the region $\mathcal{B}$ is called the *buying region* because the investor will immediately buy enough share of the underlying stock so that her portfolio will lie on $\partial\mathcal{B}$.

Moreover, from the derivation of the HJB equation (12.15), we can see that if the investor's portfolio lies in the region $\mathcal{NT}$, she would make no transaction, and for this reason, $\mathcal{NT}$ is called the *no-transaction region*.

Recall that for any constant $\rho > 0$, we have

$$U(t, S, \rho y, \rho W) = \rho^\gamma U(t, S, y, W),$$

and thus

$$\begin{aligned} \frac{\partial U}{\partial y}(t, S, \rho y, \rho W) &= \rho^{\gamma-1} \frac{\partial U}{\partial y}(t, S, y, W), \\ \frac{\partial U}{\partial W}(t, S, \rho y, \rho W) &= \rho^{\gamma-1} \frac{\partial U}{\partial W}(t, S, y, W). \end{aligned}$$

As a consequence, we can conclude that if (S, y, W) is in one of the regions, say $\mathcal{S}$, then $(S, \rho y, \rho W)$ is also in that region. In other words, each of the regions takes the form of a wedge in the (y, W) variables for any fixed S . See Figure 12.2.

It is interesting to compare this result with the case when there are no transaction costs, where it is optimal for the investor to continuously rebalance her portfolio so it will always lie on the Merton line. When transaction costs exist, it will be costly to re-balance her portfolio constantly, and as a result, it is optimal for her to re-balance her portfolio only if it drifts too far from the Merton line.

3.2 When there is a short position in option

Now we consider the case when there is a short position of a European call in the investor's portfolio at the beginning. Similar to Section 2, we

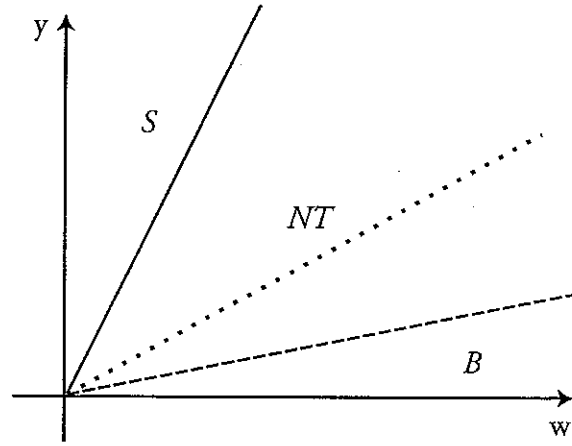


Figure 12.2. The case with transaction costs

use $\bar{U}$, $\bar{W}_T$ to denote the investor's indirect utility and terminal wealth in this case. In other words,

$$\bar{W}_T = W_T - (S_T - K)^+,$$

and

$$\bar{U}(t, S, y, W) = E_t[u(\bar{W}_T) | S_t = S, W_t = W, y_t = y].$$

The HJB equation for $\bar{U}$ is still the same as given in (12.15). However, the boundary condition is now given by

$$\bar{U}(T, S, y, W) = u(W - (S - K)^+).$$

When no option currently exist in the investor's portfolio at time 0, she can either synthetically create the option she wants, or by paying/receiving a certain price at time 0 to acquire/sell it. Assume that the investor is willing to hold a short position of a call option with maturity T and strike K for receiving a premium C upfront. The utility for the investor will be the following:

- 1 $U(0, S, 0, W)$ if she does not take the short position in call option;
- 2 $\bar{U}(0, S, 0, W + C)$ if she does take the short position in call option.

The price C for the investor to sell the call option should satisfy

$$\bar{U}(0, S, 0, W + C) \geq U(0, S, 0, W).$$

We define the *asking price* for this investor as

$$(12.16) \quad \hat{C} \equiv \min\{C \geq 0 : \bar{U}(0, S, 0, W + C) \geq U(0, S, 0, W)\}.$$

In other words, the asking price is the minimum premium required for the investor to take the liability position in the call option.

We shall point out that $\hat{C}$ may not be the option price that the option is actually traded at since the latter requires some equilibrium in the market place.

3.3 The solution to the HJB equation

To show that the indirect utility function indeed satisfies the HJB equation (12.15), let us first assume that the indirect utility function $U(t, S, y, W)$ (for simplicity of notations, we write U instead of $\bar{U}$) is a smooth function with respect to the variables t, y, S and W . In other words, U is continuously differentiable with respect to the variables t, y and W , and twice continuously differentiable with respect to the variable S .

Now we show that U satisfies (12.15). For any admissible investment/hedging strategy (L, U) , we have

$$\begin{cases} dW_t = rW_t dt - (1 + \lambda)S_t dL_t + (1 - \mu)S_t dU_t, \\ dS_t = \alpha S_t dt + \sigma S_t dz_t, \\ dy_t = dL_t - dU_t. \end{cases}$$

Applying Ito's formula (see Protter (1995), Theorem 33), we have, for any $s > t$,

$$\begin{aligned} U(s, S_s, y_s, W_s) &= U(t, S, y, W) \\ (12.17) \quad &+ \int_t^s \left(\frac{\partial}{\partial t} + \mathcal{L} \right) U(\theta, S_\theta, y_\theta, W_\theta) d\theta \\ (12.18) \quad &+ \int_t^s \left[\frac{\partial U}{\partial y} - (1 + \lambda)S_\theta \frac{\partial U}{\partial W} \right] dL_\theta \\ (12.19) \quad &+ \int_t^s \left[-\frac{\partial U}{\partial y} + (1 - \mu)S_\theta \frac{\partial U}{\partial W} \right] dU_\theta \\ (12.20) \quad &+ \int_t^s \frac{\partial U}{\partial S} \sigma S_\theta dz_\theta \\ &+ \sum_{t \leq \theta < s} [U(\theta, S_\theta, y_\theta, W_\theta) - U(\theta, S_\theta, y_{\theta-}, W_{\theta-}) \\ (12.21) \quad &- \nabla U(\theta, S_\theta, y_{\theta-}, W_{\theta-}) \cdot (\Delta y_\theta, \Delta W_\theta)], \end{aligned}$$

where

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}\sigma^2 S^2 \frac{\partial^2}{\partial S^2} + \alpha S \frac{\partial}{\partial S} + rW \frac{\partial}{\partial W}, \\ \Delta y_\theta &= y_\theta - y_{\theta-}, \\ \Delta W_\theta &= W_\theta - W_{\theta-}, \\ \nabla U &= \left(\frac{\partial U}{\partial y}, \frac{\partial U}{\partial W} \right).\end{aligned}$$

It is obvious that the investor will never buy and sell at the same time when transaction costs are incurred. In other words, the processes L and M do not have jumps at the same time. For this reason, we can consider the following two cases separately:

- When L has a jump h at time θ , i.e., the investor purchase h shares of the stock at time θ , we have

$$\begin{aligned}\Delta W_\theta &= W_{\theta-} - (1 + \lambda)S_\theta h, \\ \Delta y_\theta &= y_{\theta-} + h.\end{aligned}$$

In other words, when L has a jump, the processes (y, W) move along the direction of

$$(1, -(1 + \lambda)S),$$

which, as we have pointed earlier, implies that

$$\begin{aligned}U(\theta, S_\theta, y_\theta, W_\theta) - U(\theta, S_\theta, y_{\theta-}, W_{\theta-}) \\ \leq \frac{\partial U}{\partial y}(\theta, S_\theta, y_{\theta-}, W_{\theta-})h - (1 + \lambda)S \frac{\partial U}{\partial W}(\theta, S_\theta, y_{\theta-}, W_{\theta-})h \\ = \nabla U(\theta, S_\theta, y_{\theta-}, W_{\theta-}) \cdot (\Delta y_\theta, \Delta W_\theta);\end{aligned}$$

- Similarly, we can show that the above inequality holds when M has a jump.

As a result, the summation in (12.21) is non-positive.

Moreover, because the function U satisfies the equation (12.15), the integrands in (12.17), (12.18) and (12.19) are all non-positive. Noticing that the integral in (12.20) is a martingale, we have

$$EU(s, S_s, y_s, W_s) \leq U(t, S, y, W).$$

Taking $s = T$, it follows that

$$(12.22) \quad U(t, S, y, w) \geq EU(T, S_T, y_T, W_T) = Eu(W_T).$$

On the other hand, given the function U that satisfies the equation (12.15), we can define the regions $\mathcal{S}$ and $\mathcal{B}$ as before. Introduce the following processes

$$(12.23) \quad dW_t = rW_t dt - (1 + \lambda)S_t dL_t + (1 - \mu)S_t dM_t,$$

$$(12.24) \quad dy_t = dL_t - dM_t,$$

$$(12.25) \quad L_t = \int_0^t 1_{\{(y_s, S_s, W_s) \in \partial \mathcal{B}\}} dL_s,$$

$$(12.26) \quad M_t = \int_0^t 1_{\{(y_s, S_s, W_s) \in \partial \mathcal{S}\}} dM_s.$$

With similar arguments as before, we can show that with this strategy, we have

$$U(t, S, y, W) = E_t u(W_T).$$

Therefore

$$U(t, S, y, W) = \max_{(L, M)} E_t u(W_T).$$

In other words, the function U is indeed the indirect utility function.

3.4 Viscosity solution to the HJB equation

We have assumed that the indirect utility function U is smooth in its variables. However, this may not be the case when there are transaction costs to the investment problem. For this reason, the concept of *viscosity solution* is needed.

We first define a functional (or operator) F on the smooth functions f :

$$F(f) = \max \left\{ \frac{\partial f}{\partial y} - (1 + \lambda)S \frac{\partial f}{\partial W}, -\frac{\partial f}{\partial y} + (1 - \mu)S \frac{\partial f}{\partial W}, \right. \\ \left. \frac{\partial f}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} + \alpha S \frac{\partial f}{\partial S} + rW \frac{\partial f}{\partial W} \right\}.$$

Note that $F(f)$ is also a function of the variables (t, S, y, W) .

A continuous function U is a *viscosity solution* to the HJB equation (12.15) if it satisfies the following conditions:

- 1 For any smooth function ϕ , if (t_0, S_0, y_0, W_0) is a local maximum point of the function $U - \phi$, then

$$F(\phi)(t_0, S_0, y_0, W_0) \geq 0.$$

- 2 For any smooth function ϕ , if (t_0, S_0, y_0, W_0) is a local minimum point of the function $U - \phi$, then

$$F(\phi)(t_0, S_0, y_0, W_0) \leq 0.$$

The concept of viscosity solution is primarily used in optimal control problems because their value functions are usually not smooth. For details on the concept of viscosity solutions, see Crandall, et al (1983). For its applications to optimal control problems, especially to those problems similar to the one considered in this paper, see Haussmann and Suo (1995).

It can be shown rigorously that the indirect utility function U is indeed the *unique* viscosity solution to the HJB equation (12.15). See Haussmann and Suo (1995) for the proof of this result under more general conditions.

4. Numerical method for solving the HJB equation

It is very difficult, if not impossible, to derive the analytical solution to the equation (12.15). For this reason, we propose a numerical method to compute the indirect utility function.

For any $t > 0$ and $s > t$, it can be shown that the following equation holds (usually called the dynamic programming principle, see Haussmann and Suo (1995) for a proof):

$$(12.27) \quad U(t, S, y, W) = \max_{(L, M)} E_t[U(s, S_s, y_s, W_s)],$$

where the expectation is taken conditional on $(S_t = S, y_t = y, W_t = W)$.

Recall that the stock price follows a geometric Brownian motion. In order to compute the indirect utility function numerically, we first divide the time interval $[0, T]$ into N subintervals:

$$0 < \Delta_t < 2\Delta_t < \dots < N\Delta_t = T,$$

where $\Delta_t = T/N$. We can approximate the random factor, which is represented by a Brownian motion, z by a binomial process z^* :

$$z^*(i+1) = \begin{cases} z^*(i) \cdot k_u, & \text{with probability } 1/2, \\ z^*(i) \cdot k_d, & \text{with probability } 1/2, \end{cases}$$

where k_u and k_d are determined such that the expected value and variance of the Brownian motion z on $[i\Delta_t, (i+1)\Delta_t]$ are matched. It follows that

$$k_u = \sqrt{\Delta_t}, \quad k_d = -\sqrt{\Delta_t}.$$

The stock price S is approximated by the binomial process

$$S^*(i+1) = \begin{cases} S^*(i) \cdot \exp(\alpha\Delta_t + \sigma\sqrt{\Delta_t}), & \text{with probability } 1/2, \\ S^*(i) \cdot \exp(\alpha\Delta_t - \sigma\sqrt{\Delta_t}), & \text{with probability } 1/2. \end{cases}$$

We will also assume that the minimum amount of stocks one can buy/sell in each transaction is Δ_y . The y -axis can then be divided into

$$-M\Delta_y < \dots < -\Delta_y < 0 < \Delta_y < \dots < M\Delta_y,$$

where M is chosen to be large enough so that the regions outside the interval $[-M\Delta_y, M\Delta_y]$ are economically irrelevant.

From (12.27), and the form of the optimal investment/hedging strategy described in (12.23)-(12.26), we can write,³ for any t ,

$$\begin{aligned} U^\Delta(t, S, y, W) &= \max_{(L, M)} E_t(U^\Delta(t + \Delta_t, S_{t+\Delta_t}, y_{t+\Delta_t}, W_{t+\Delta_t})) \\ &= \max \left\{ U^\Delta(t, S, y + \Delta_y, W - (1 + \lambda)S\Delta_y), \right. \\ &\quad U^\Delta(t, S, y - \Delta_y, W + (1 - \mu)S\Delta_y), \\ &\quad \left. E(U^\Delta(t + \Delta_t, S_{t+\Delta_t}, y, \exp(r\Delta_t)W)) \right\}. \end{aligned} \quad (12.28)$$

In other words, the optimal decision is described by three different actions:

- 1 sell minimum number of shares allowed;
- 2 buy minimum number of shares allowed; and
- 3 make no transactions.

As discussed earlier, when the portfolio falls in the sell region S , it is optimal for the investor to sell enough (or the minimum number of) shares so that the portfolio will fall on the boundary of the no-transaction region. The opposite action should be taken when the portfolio falls in the buy region B . When the portfolio falls in the region NT , it is optimal for the investor to make no transactions.

To be more specific, no transaction will be made if (for clarity of notations, we drop Δ from the approximate indirect utility function U^Δ)

$$\begin{aligned} \frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W} &< 0, \\ -\frac{\partial U}{\partial y} + (1 - \mu)S \frac{\partial U}{\partial W} &< 0. \end{aligned}$$

In this case,

$$\begin{aligned}
 U(t, S, y, W) &= E_t[U(t + \Delta_t, S_{t+\Delta_t}, y_{t+\Delta_t}, W_{t+\Delta_t})] \\
 &= \frac{1}{2} U(t + \Delta_t, S \exp((\alpha - r)\Delta_t + \sigma\sqrt{\Delta_t}), \\
 &\quad y_{t+\Delta_t}, \exp(r\Delta_t)W) \\
 &\quad + \frac{1}{2} U(t + \Delta_t, S \exp((\alpha - r)\Delta_t - \sigma\sqrt{\Delta_t}), \\
 &\quad y_{t+\Delta_t}, \exp(r\Delta_t)W).
 \end{aligned}$$

The action to buy a minimum number of shares will be taken if

$$\frac{\partial U}{\partial y} - (1 + \lambda)S \frac{\partial U}{\partial W} = 0,$$

in which case

$$U(t, S, y, W) = U(t, S, y + \Delta_y, W - (1 + \lambda)S\Delta_y).$$

The action to sell a minimum number of shares will be taken if

$$-\frac{\partial U}{\partial y} + (1 - \mu)S \frac{\partial U}{\partial W} = 0,$$

in which case

$$U(t, S, y, W) = U(t, S, y - \Delta_y, W + (1 - \mu)S\Delta_y).$$

Based on these results, we now propose a numerical scheme to compute the indirect utility function U :

- Divide the time interval $[0, T]$ and the y -axis as before. The W -axis can also be divided into K -intervals:

$$0 < \Delta_W < \dots < K\Delta_W.$$

- Approximate the Brownian motion z by the binomial process z^* , and define, for (i, j, k, l) ,

$$u(i, j, k, l) \equiv U(i\Delta_t, Sk_u^j k_d^{i-j}, k\Delta_y, l\Delta_W),$$

where $0 \leq i \leq N, 0 \leq j \leq i, -M \leq k \leq M$, and $0 \leq l \leq K$.

- At the maturity time T ,

$$u(N, j, k, l) = l\Delta_W - (Sk_u^j k_d^{N-j} - K)^+$$

for all (j, k, l) , where S is the stock price at $t = 0$.

- Recursively, for $i < N$, define

$$u^*(i, j, k, l) = \frac{1}{2}[u(i+1, j+1, k, l^+) + u(i+1, j, k, l^+)]$$

where l^+ is the integer such that corresponding mesh point for W is closest to the value $\exp(r\Delta_t)l\Delta_W$.⁴

- Check if the following relationships hold:

$$\begin{aligned} u^*(i, j, k, l) &> u^*(i, j, k+1, l^+), \\ u^*(i, j, k, l) &> u^*(i, j, k-1, l^-), \end{aligned}$$

where l^+ (l^-) is the integer such that corresponding mesh point for W is closest to the value $l\Delta_W - (1+\lambda)S\Delta_y$ ($l\Delta_W + (1-\mu)S\Delta_y$, respectively). Such mesh points fall in the no-transaction region, so we can let

$$u(i, j, k, l) = u^*(i, j, k, l).$$

- Take such a point, say (j_0, k_0, l_0) . Notice that U is a non-decreasing function in the variables y and W . Working recursively, we can check if

$$u^*(i, j, k, l) \leq u^*(i, j, k+1, l^+),$$

in which case, we can set

$$u(i, j, k, l) = u^*(i, j, k+1, l^+).$$

Similarly, if

$$u^*(i, j, k, l) \leq u^*(i, j, k-1, l^-),$$

we can set

$$u(i, j, k, l) = u^*(i, j, k-1, l^-).$$

In other words, we will make sure that for all (j, k, l) ,

$$\begin{aligned} u(i, j, k, l) &\geq u(i, j, k+1, l^+), \\ u(i, j, k, l) &\geq u(i, j, k-1, l^-). \end{aligned}$$

This scheme can be easily implemented, and the indirect utility function can thus be computed numerically. With the indirect utility, we can study the optimal hedging strategy by studying the no-transaction region.

Similarly, the indirect utility for the investment problem without option can be computed numerically by changing the value of the function U at time T . The asking price of the option can thus be obtained through (12.16).

5. Summary

In this paper, we study the optimal investment problem for an investor with a HARA type utility function. We show that when there are no transaction costs, and there is a short position in the investor's portfolio, she may prefer not to perfectly hedge the option, as in the Black-Scholes option pricing framework. In fact, when the growth rate of the stock is greater than the risk-free rate, the investor will always "over" hedge the option. When there is transaction cost in the investor's portfolio, the investor's optimal investment/hedging strategy can be described by three regions: the buying region, the selling region and the no transaction region. When her portfolio falls in the buying region, she will buy enough shares of the stock to make her portfolio lie in the no transaction region (NT). Similar results hold when her portfolio falls in the selling region. When her portfolio falls in the NT region, it is optimal for the investor to make no transaction, i.e., simply hold her portfolio. Viscosity solution is introduced to describe the indirect utility function. We also propose a numerical method to compute the indirect utility function, and the asking price for an option for be computed as a result.

The numerical scheme we proposed in this paper remains to be implemented in order to show numerically the properties of the hedging strategy.

6. Appendix

Solving the HJB equation with HARA utility function:

We guess that the indirect utility function, or the solution to (12.5) is in the following form:

$$U(t, W) = g(t) \frac{W^\gamma}{\gamma}.$$

Then

$$\begin{aligned} U_t &= g'(t) \frac{W^\gamma}{\gamma}, \\ U_W &= g(t) W^{\gamma-1}, \\ U_{WW} &= g(t) (\gamma-1) W^{\gamma-2}, \end{aligned}$$

and

$$\pi^* = -\frac{\alpha - r}{\sigma^2} \frac{U_W}{W U_{WW}} = \frac{\alpha - r}{\sigma^2 (1 - \gamma)}.$$

Substitute these expressions into the HJB equation (12.5), we get

$$g'(t) + \nu\gamma g(t) = 0,$$

where

$$\nu \equiv \frac{1}{2} \left[\left(\frac{\alpha - r}{\sigma} \right)^2 + \frac{1}{1 - \gamma} + r \right].$$

The boundary condition reduces to

$$g(T) = 1.$$

The indirect utility can now be obtained by solving this equation.

Proof of (12.10):

In the case when there is no transaction cost, it is well-known that the call option can be replicated by a self-financing portfolio strategy that requires an initial cash of

$$c = SN(d_1) - e^{-r(T-t_0)}KN(d_2).$$

We denote this strategy by π^c . We will show that if we denote by $\bar{U}(t, S, W)$ as the indirect utility function for the investor at time t when her wealth is W and the current (or spot) stock price is S , then

$$\bar{U}(0, S, W) = U(0, W - c).$$

In fact, if this is not the case, say,

$$\bar{U}(0, S, W) < U(0, W - c),$$

then let π_t be the optimal portfolio that requires an investment of $W - c$ at the beginning when there is no short position in call in her portfolio. By definition,

$$U(0, W - c) = EU(T, W_T^\pi),$$

Consider the new trading

$$\bar{\pi} = \pi + \pi^0,$$

which needs an initial investment of

$$(W - c) + c = W,$$

and has the terminal wealth of

$$W_T^\pi + (S_T - K)^+.$$

Thus we have

$$\begin{aligned}
 \bar{U}(0, S, W) &\geq EU(T, S_T, W_T^\pi) \\
 &= EU(T, S_T, W_T^\pi + (S_T - K)^+ - (S_T - K)^+) \\
 &= EU(T, S_T, W_T^\pi) \\
 &= E\left(\frac{(W_T^\pi)^\gamma}{\gamma}\right) \\
 &\geq U(0, W - c),
 \end{aligned}$$

which is a contradiction. In other words, we must have

$$\bar{U}(0, S, W) \geq U(0, W - c).$$

The other side of the inequality can be shown similarly.

In general, we can show that

$$\bar{U}(t, S, W) = U(W - f(t, S)),$$

where S is the spot price of the stock at time t , and $f(t, S)$ is the price of the call option when the spot stock price is S .

Notes

1. It is obvious that this result does not hold for general utility functions.
2. In the optimal control terminology, this type of problems are called *singular control* problems because the processes L and M may be singular functions, i.e., they may be continuous and yet non-differentiable.
3. It can be shown the function defined this way converges to the indirect utility function as $N \rightarrow \infty, \Delta t, \Delta y \rightarrow 0$, see Barles and Souganidis (1991) for the proof of a similar result.
4. For better approximation, we can use interpolations at $(i+1)\Delta t$ for this value

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